

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Document Camera
Manufacturer: Elmo
Firmware Version: N/A
Model(s): P30S

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.08.0002-b002	1.4.2

Version History

Module Version	Date	Notes
1_3_0_0	7/30/2018	Fixed status feedback. Removed Light On command. Added Base and Upper state for Light command.
1_1_1_1	6/21/2018	Updated module script to conform to latest standards
1_1_1_0	2/12/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='P30S')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='P30S')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AutoFocus	Value None		
# AutoFocus example InterfaceName.Set('AutoFocus', None)			
Command Color	Value 'Color'	Value 'B/W'	
# Color example InterfaceName.Set('Color', 'Color')			
Command Display	Value 'Single'	Value '3x3'	Value '4x4'
# Display example InterfaceName.Set('Display', 'Single')			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command Focus	Value 'Near'	Value 'Far'	Value 'Stop'
# Focus example InterfaceName.Set('Focus', 'Near')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Gamma	Value 'Normal'	Value 'High'	Value 'Low'
# Gamma example InterfaceName.Set('Gamma', 'Normal')			
Command ImageRotation	Value '0'	Value '180'	
# ImageRotation example InterfaceName.Set('ImageRotation', '0')			
Command ImageSave	Value None		
# ImageSave example			

Global Scripter Module Communication Sheet

InterfaceName.Set('ImageSave', None)			
Command Input	Value 'RGB'	Value 'SD'	Value 'Camera'
# Input example InterfaceName.Set('Input', 'RGB')			
Command Iris	Value 'Open' 'Auto'	Value 'Close' 'Manual'	Value 'Stop'
# Iris example InterfaceName.Set('Iris', 'Open')			
Command Light	Value 'Base'	Value 'Upper'	Value 'Off'
# Light example InterfaceName.Set('Light', 'Base')			
Command PosNeg	Value 'Pos'	Value 'Neg'	
# PosNeg example InterfaceName.Set('PosNeg', 'Pos')			
Command SlideShow	Value 'Start'	Value 'Stop'	
# SlideShow example InterfaceName.Set('SlideShow', 'Start')			
Command SlideShowRepeat	Value 'On'	Value 'Off'	
# SlideShowRepeat example InterfaceName.Set('SlideShowRepeat', 'On')			
Command TextEnhancer	Value 'Graphics' 'Text3'	Value 'Text1'	Value 'Text2'
# TextEnhancer example InterfaceName.Set('TextEnhancer', 'Graphics')			
Command USBMode	Value 'Application'	Value 'Storage'	
# USBMode example InterfaceName.Set('USBMode', 'Application')			
Command Zoom	Value 'Tele'	Value 'Wide'	Value 'Stop'
# Zoom example InterfaceName.Set('Zoom', 'Tele')			

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command Color	Value 'Color'	Value 'B/W'	
# Color examples InterfaceName.Update('Color') Value = InterfaceName.ReadStatus('Color') InterfaceName.SubscribeStatus('Color', None, FeedbackHandler)			
Command Display	Value 'Single'	Value '3x3'	Value '4x4'
# Display examples InterfaceName.Update('Display') Value = InterfaceName.ReadStatus('Display') InterfaceName.SubscribeStatus('Display', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Gamma	Value 'Normal'	Value 'High'	Value 'Low'
# Gamma examples InterfaceName.Update('Gamma') Value = InterfaceName.ReadStatus('Gamma') InterfaceName.SubscribeStatus('Gamma', None, FeedbackHandler)			
Command ImageRotation	Value '0' – '180'		
# ImageRotation examples			

Global Scripter Module Communication Sheet

InterfaceName.Update('ImageRotation') Value = InterfaceName.ReadStatus('ImageRotation') InterfaceName.SubscribeStatus('ImageRotation', None, FeedbackHandler)			
Command Input	Value 'RGB'	Value 'SD'	Value 'Camera'
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command Iris	Value 'Auto'	Value 'Manual'	
# Iris examples InterfaceName.Update('Iris') Value = InterfaceName.ReadStatus('Iris') InterfaceName.SubscribeStatus('Iris', None, FeedbackHandler)			
Command Light	Value 'Base'	Value 'Upper'	Value 'Off'
# Light examples InterfaceName.Update('Light') Value = InterfaceName.ReadStatus('Light') InterfaceName.SubscribeStatus('Light', None, FeedbackHandler)			
Command PosNeg	Value 'Pos'	Value 'Neg'	
# PosNeg examples InterfaceName.Update('PosNeg') Value = InterfaceName.ReadStatus('PosNeg') InterfaceName.SubscribeStatus('PosNeg', None, FeedbackHandler)			
Command TextEnhancer	Value 'Graphics' 'Text3'	Value 'Text1'	Value 'Text2'
# TextEnhancer examples InterfaceName.Update('TextEnhancer') Value = InterfaceName.ReadStatus('TextEnhancer') InterfaceName.SubscribeStatus('TextEnhancer', None, FeedbackHandler)			
Command USBMode	Value 'Application'	Value 'Storage'	
# USBMode examples InterfaceName.Update('USBMode') Value = InterfaceName.ReadStatus('USBMode') InterfaceName.SubscribeStatus('USBMode', None, FeedbackHandler)			

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

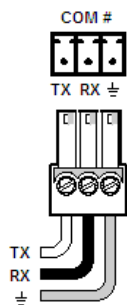
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Appendix A. Set Commands

Auto Focus None	\x02AF0 \x03
Color B/W	\x02CB1 \x03
Color Color	\x02CB0 \x03
Display 3x3	\x02DP1 \x03
Display 4x4	\x02DP2 \x03
Display Single	\x02DP0 \x03
Executive Mode Off	\x02LL0 \x03
Executive Mode On	\x02LL1 \x03
Focus Far	\x02F0- \x03
Focus Near	\x02F0+ \x03
Focus Stop	\x02F00 \x03
Freeze Off	\x02FZ0 \x03
Freeze On	\x02FZ1 \x03
Gamma High	\x02GM0 \x03
Gamma Low	\x02GM2 \x03
Gamma Normal	\x02GM1 \x03
Image Rotation 0	\x02R00 \x03
Image Rotation 180	\x02R01 \x03
Image Save None	\x02CA0 \x03
Input Camera	\x02AV0 \x03
Input RGB	\x02AV1 \x03
Input SD	\x02AV2 \x03
Iris Auto	\x02IR1 \x03
Iris Close	\x02IR- \x03
Iris Manual	\x02IR2 \x03
Iris Open	\x02IR+ \x03
Iris Stop	\x02IR0 \x03
Light Base	\x02PL1 \x03
Light Off	\x02PL0 \x03
Light Upper	\x02PL2 \x03
Pos Neg Neg	\x02NP1 \x03
Pos Neg Pos	\x02NP0 \x03
Slide Show Start	\x02SS1 \x03
Slide Show Stop	\x02SS0 \x03
Slide Show Repeat Off	\x02SR1 \x03
Slide Show Repeat On	\x02SR0 \x03

**Global Scripter Module
Communication Sheet**

Text Enhancer Graphics	\x02CT0 \x03
Text Enhancer Text1	\x02CT1 \x03
Text Enhancer Text2	\x02CT2 \x03
Text Enhancer Text3	\x02CT3 \x03
USB Mode Application	\x02UM1 \x03
USB Mode Storage	\x02UM0 \x03
Zoom Stop	\x02Z00 \x03
Zoom Tele	\x02Z0+ \x03
Zoom Wide	\x02Z0- \x03

Appendix B. Update Commands

Text Enhancer	\x02Q52 \x03
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